🔍 Filters

Risk Level

HIGH ×
MEDIUM ×
LOW ×

☐ Show Anomalies Only

☐ Show Silent Risk Only

Sex

M × F ×

Age Range

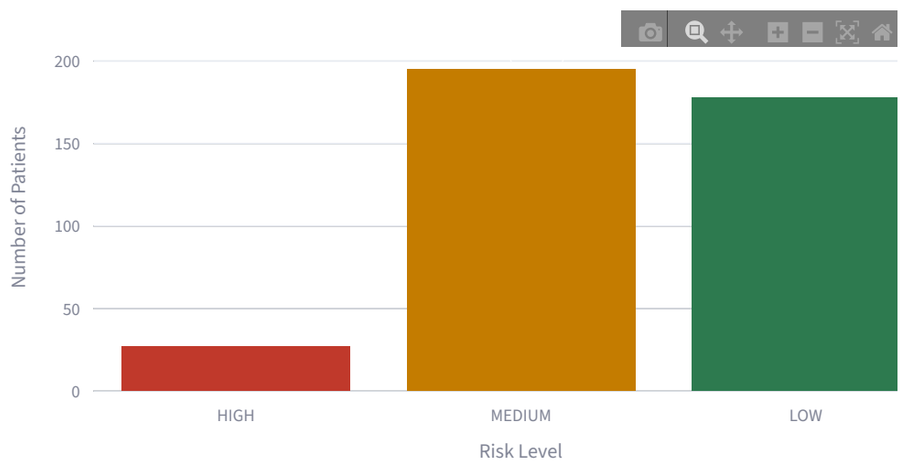
21 89

🏥 Health Risk Prediction System

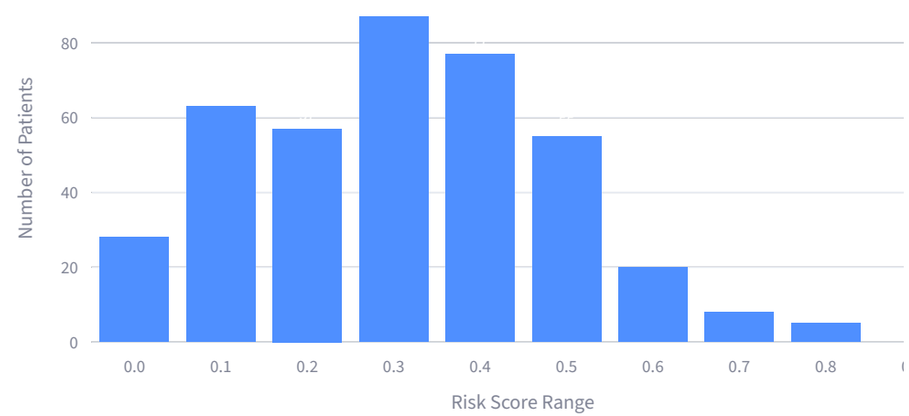
Capstone Project — Healthcare Vertical | Model: llama-3.1-8b-instant | Embeddings: all-MiniLM-L6-v2 | Vector Store: FAISS

Total Patients	HIGH Risk	MEDIUM Risk	Anomalies	Silent Risk Profiles
400	27	195	11	31
	↑ 6.75%	↑ 48.75%		

📊 Risk Distribution



📊 Risk Score Distribution



📋 Patient Risk Table — 400 records

	patient_id	age	sex	bmi	smoker	diabetic	dx_code	risk_score	risk_level	anomaly	silent_risk
0	H10000	68	M	25.3	0	0	E78	0.35	LOW	☐	☐
1	H10001	70	M	23.4	1	0	J45	0.60	MEDIUM	☐	☑
2	H10002	41	M	24.5	0	0	M54	0.20	LOW	☐	☐
3	H10003	44	M	16.2	0	1	K21	0.30	LOW	☐	☐
4	H10004	77	F	33.3	0	0	F41	0.50	MEDIUM	☐	☐
5	H10005	53	F	28.1	0	0	G47	0.25	LOW	☐	☐
6	H10006	46	F	26.3	0	0	I10	0.35	LOW	☐	☐
7	H10007	80	F	23.5	0	0	I10	0.50	MEDIUM	☐	☐
8	H10008	24	M	28.6	0	0	G47	0.15	LOW	☐	☐
9	H10009	67	M	31.4	1	1	E11	0.90	HIGH	☑	☐

🤖 LLM-Generated Risk Summary

Select Patient ID

H10009

Patient Profile

Age: 67

Sex: M

BMI: 31.4

LLM-Generated Business Summary

Patient H10009 is at high risk for potential health complications and has a high chance of increasing healthcare costs due to his age, BMI, smoking status, and diabetes. Historically, similar patients with similar characteristics have seen an increase in health issues, suggesting that this patient could experience similar health challenges unless proactive steps are taken. Our analysis recommends implementing preventive measures and close monitoring to mitigate this risk and

Smoker: Yes

Diabetic: Yes

DX Code: E11

Risk Score: 0.9

Risk Level: HIGH

⚠️ ANOMALY FLAGGED

ensure optimal patient outcomes.

⚠️ Anomaly Report

	patient_id	age	sex	bmi	smoker	diabetic	dx_code	risk_score	risk_level	anomaly	silent_risk
9	H10009	67	M	31.4	1	1	E11	0.90	HIGH	☑	☐
12	H10012	72	M	28.4	1	1	J45	0.85	HIGH	☑	☑
33	H10033	74	M	24.2	1	1	E11	0.90	HIGH	☑	☐
51	H10051	89	M	31.6	1	1	G47	0.90	HIGH	☑	☑
52	H10052	86	F	13.1	0	0	M54	0.55	MEDIUM	☑	☐
192	H10192	54	M	39.8	0	1	G47	0.60	MEDIUM	☑	☑
292	H10292	70	M	35.5	0	1	E11	0.90	HIGH	☑	☐
300	H10300	43	M	38.7	0	0	I25	0.50	MEDIUM	☑	☐
315	H10315	41	M	14.8	0	0	E78	0.35	LOW	☑	☐
320	H10320	44	F	39.7	0	0	I25	0.50	MEDIUM	☑	☐

🔕 Silent Risk Profiles

	patient_id	age	sex	bmi	smoker	diabetic	dx_code	risk_score	risk_level	anomaly	silent_risk
120	H10120	71	F	31.2	1	0	F41	0.70	HIGH	☐	☑
125	H10125	80	F	23.2	1	0	M54	0.60	MEDIUM	☐	☑
127	H10127	65	F	25.9	1	1	F41	0.75	HIGH	☐	☑
173	H10173	63	M	32.5	0	1	E78	0.60	MEDIUM	☐	☑
177	H10177	61	F	33.4	1	0	E78	0.60	MEDIUM	☐	☑
181	H10181	87	M	22.4	1	0	F41	0.60	MEDIUM	☐	☑
184	H10184	51	F	30.3	1	1	J45	0.70	HIGH	☐	☑
192	H10192	54	M	39.8	0	1	G47	0.60	MEDIUM	☑	☑
213	H10213	70	M	22.7	1	0	J45	0.60	MEDIUM	☐	☑
227	H10227	78	M	38.1	0	0	J45	0.60	MEDIUM	☐	☑

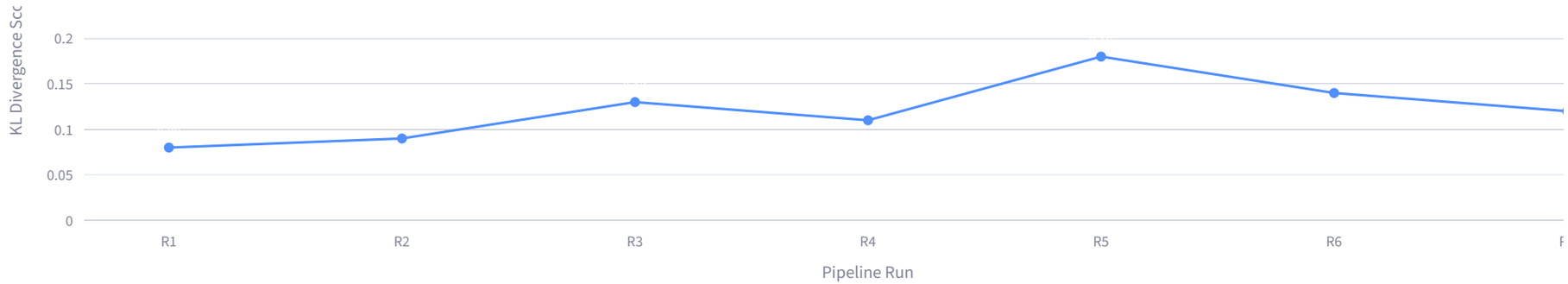
📡 LLMOps Pipeline Summary

Model
llama-3.1-8b-instant

Avg Latency
2.88s

Avg Tokens
91

Embedding Drift
0.12 — STABLE



Drift threshold: 0.25 — Retraining triggered if exceeded for 2 consecutive runs

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Health Risk Prediction System | Capstone Project — Saravanan Shanmugam 207681 | Generative AI Advanced